

PERRY LIN

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EDUCATION

University of California, Santa Cruz

September 2020 - June 2024

BS in Computer Science GPA: 3.75

Santa Cruz, CA

- *Relevant Coursework:* Data Structures and Algorithms, Distributed Systems, Web Applications, Computational Models, Machine Learning, Foundation of Programming Languages, Algorithm Analysis, Computer Security, Computer Architecture

PROJECTS

GenAI Recipe Companion | *React Native, Django, PostgreSQL, Google APIs, OpenAI API*

- Designed and developed a comprehensive recipe application integrating Google APIs, offering users access to an extensive database of culinary options and ingredient listings.
- Implemented AI-driven ingredient substitution suggestions using the OpenAI API, personalizing recommendations based on user preferences, dietary restrictions, and ingredient availability.
- Integrated geolocation features with Google Maps API to help users locate and compare nearby grocery stores by price and proximity, enhancing shopping convenience.
- Ensured scalability and high performance by utilizing React Native for cross-platform development, Django for backend services, and PostgreSQL for reliable data storage.

Project and Task Organizer | *Python, React, JavaScript, HTML, OpenAI API*

- Led a development team to design and build a comprehensive task management platform enabling users to create, assign, and collaboratively track tasks in real-time.
- Implemented a project module allowing users to create projects containing tasks, enhancing organization by providing a hierarchical structure for task management.
- Created an intuitive task search and filtering feature, allowing users to efficiently sort and manage tasks according to various parameters.
- Integrated generative AI features with the OpenAI API, allowing users to prompt Groq to generate project ideas and tasks, innovating the brainstorming process.
- Deployed the application to a class of 40 high school students, streamlining coordination and promoting collaborative work, resulting in improved project completion rates.

Distributed Storage System | *Python, Flask, Docker*

- Co-developed a sharded, replicated, and fault-tolerant key-value store to ensure high availability and scalability in distributed systems.
- Optimized data management across distributed nodes by implementing RESTful APIs for shard member retrieval and shard key-counts.
- Implemented dynamic shard operations—including shard ID retrieval, member addition, and resharding—through GET and PUT requests, enhancing system flexibility.
- Improved system resilience by integrating replication and fault tolerance mechanisms, ensuring consistent operation under node failures and distributed workloads.

Serverless URL Shortener | *AWS Lambda, DynamoDB, Python, HTML*

- Developed a scalable, serverless URL shortening service using AWS Lambda and DynamoDB, generating unique 6-character codes for efficient redirects.
- Designed a RESTful API with AWS API Gateway to handle high volumes of URL requests and redirections, supporting both JSON and form data inputs.
- Enhanced user accessibility by creating an interactive front-end interface with HTML and JavaScript, allowing users to input URLs and receive shortened links effortlessly.

SKILLS

Languages:	Python(Flask, Django), JavaScript, HTML/CSS, C, C++, Java, TypeScript, Vue.js, Node.js
Software/Tools:	AWS(Lambda, DynamoDB), Docker, Git, Postman, Virtual Studio, MySQL, PostgreSQL